Reflexión: cadenas de Markov

nota: preferí hacer la reseña en inglés porque me resultó más intuitivo después de haber visto los videos en ese idioma.

First video:

First and foremost, I really enjoyed this video. It was really fun and easy to digest, 5/5 stars, I will recommend it.

Now, onto the actual content itself, “The Strange Math that Predicts (Almost) Anything” explains how probability evolved from simple independent events to more realistic systems in which events depend on previous outcomes. It starts with the law of large numbers, showing that when we repeat many trials, averages tend to approach the expected value. From there, the video introduces Markov chains, which describe systems where the next state depends mainly on the current one rather than the full past. This idea is powerful because it simplifies very complex phenomena while still allowing us to make actually useful predictions. The video shows how this logic can be applied to language, games like solitaire, nuclear physics, internet search, and predictive text (ofc other mathematicians were the ones that applied it). One especially interesting point is that the web itself can be modeled as a Markov chain, which helped create better ways to rank pages by quality instead of just counting how many results could be returned.

What I found most important is that the video shows probability not as abstract math, but as a practical tool for dealing with uncertainty in real-world events. I finally understood that although at first glance the “memoryless” nature of Markov chains sounded more like a limitation, in reality it is actually what makes them mathematically manageable and broadly useful. The card-shuffling example also reinforces this idea: randomness can be studied precisely, and for a standard 52-card deck, the famous result is that about seven riffle shuffles are enough to mix it well.

Second video:

The second video, “Hidden Markov Model: Data Science Concepts”, builds on that foundation by introducing situations where the real state of a system cannot be observed directly. Although simple in its setting, I also really liked this setting, Ritvik talked at the perfect pace.

So, in a Hidden Markov Model, we observe outputs, but the underlying state remains hidden. In the example, the visible shirt color is the observation, while the teacher’s mood is the hidden state. The model is built around two main probabilities: **transition probabilities**, which describe how one hidden state changes into another, and **emission probabilities**, which describe how likely each observation is given a hidden state.

The most important idea in this video is that machine learning often has to work with **indirect evidence** rather than direct knowledge. We usually cannot observe the true cause of something, only the signals it produces. Hidden Markov Models are valuable because they provide a structured way to infer those hidden causes from observable data. I can’t believe I have been taking a Machine Learning class and this is the first time I'm understanding this…..